MUHAMMED FAIYAZUDDEEN

FULL STACK SOFTWARE ENGINEER | SOLUTION ARCHITECT
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SUMMARY

Full stack software engineer and solution architect with 11+ years delivering high performance, scalable web applications across government, healthcare, fintech, and public transit. Strengths include performance optimization, real time systems (WebSockets), PWAs/Service Workers, RxJS reactive pipelines, and distributed architectures. Managed teams of up to 10 engineers and led remote delivery for international clients, consistently driving substantial improvements in speed, reliability, and throughput. Experienced in corporate and non profit full stack development training. Currently providing independent consultation (Feb 2026 – Present).

CORE COMPETENCIES

- Performance optimization
- Real time systems
- PWAs/Service Workers
- RxJS reactive pipelines
- Distributed architectures
- Team management
- Remote delivery
- Full stack development

TECHNICAL TOOLKIT

Frontend & PWA: React • Next.js • Angular • Redux • RxJS • Service Workers • PWA
Backend & Systems: Node.js • Express.js • Kotlin • Java (Spring Boot) • Python
Programming: JavaScript (ES6+) • TypeScript • Kotlin • Java • Python
Databases: PostgreSQL • MySQL • MongoDB • Elasticsearch • IndexedDB • Firebase
Analytics: Datadog

EXPERIENCE

04/205 – 02/2026

Product Development Engineer | Robogebra

- Built a Retrieval Augmented Generation (RAG) pipeline using LangChain, ChromaDB, and OpenAI embeddings to reduce LLM hallucination and ground responses in proprietary knowledge bases.
- Engineered real time LLM streaming with Server-Sent Events (SSE), delivering token by token responses and enabling live tool execution visibility.
- Implemented agentic tool calling with LangChain and Zod schema validation, bridging natural language intent to structured API actions.
- Designed a PDF ingestion pipeline using LangChain for semantic search across uploaded knowledge bases.
- Built a hybrid semantic search pipeline combining MongoDB Atlas $vectorSearch with fuzzy keyword parsing to resolve structured student queries.
- Developed a multi language LLM prompt orchestration system supporting six response modes (English, Tamil, Hindi, Tanglish, Hinglish, bilingual script+English math).
- Built a 7-stage LLM response post processing pipeline to safely render KaTeX/LaT in the browser, reducing malformed rendering errors in production.

09/2014 – 01/2025

Solution Architect / Technical Lead | Provility

- Led technical delivery across 8+ international client engagements; managed and mentored teams of 5–10 engineers and owned end-to-end architecture, delivery, and operational readiness.
- Architected Service Worker-based offline interceptors for session continuity in constrained environments.
- Rebuilt complex client-side engines (SurveyJS 3,500-question evaluation) with batching, async microtask deferral, and caching.
- Implemented WebSocket-based real-time features (conflict detection, driver tracking, call-state sync).

- Introduced performance budgets and regression test practices, achieving 40–90% improvements in load and interaction times.
- Migrated 50+ legacy components to React with virtualization and accessibility improvements.
- Built RxJS-based reactive pipelines for geospatial and market data with client-side caching.
- Authored Cypress E2E suites covering critical flows and complex map/transit interactions.
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Selected Client Engagements (under Provility):

- Averq – Unified Compliance & Audit Platform (11/2024 – 01/2026): Rebuilt SurveyJS engine; improved form load from 14s to 4s (71%, measured by custom benchmark). Architected offline interceptors and WebSocket conflict detection.
- Nexum – Account-Based Ticketing Platform (02/2023 – 04/2024): Delivered Kotlin coroutine APIs and Java GTFS cache; integrated Datadog + Slack alerting.
- SagaMath – Interactive Math Worksheet Platform (01/2021 – 12/2021): Built safe client-side JS execution engine; KaTeX rendering and live worksheet authoring.
- AdeptIQ – Paratransit Dispatch Engine (07/2019 – 01/2021): Embedded React data grid; designed booking service model; implemented real-time driver tracking.
- Tradeworks – Algorithmic Trading Automation Platform (01/2019 – 03/2019): Built RxJS pipelines for forex streaming with back-pressure.
- Eform – Government E-Application Platform (03/2018 – 11/2018): Built no-code service designer; improved drag-and-drop performance by >85%.
- Choice Health Brokers – Internal CRM Platform (05/2017 – 01/2018): Integrated Twilio Voice API; architected Redux-driven call-state sync.
- BigZeta – Declarative Data Visualization Platform (01/2017 – 04/2017): Built reusable React component library for charting.
- WeldQ – Welder Qualification Certification Platform (01/2016 – 01/2017): Enginee online exam engine with auto-grading and dashboards.
- WeldTrace – Smart Welding Job Management Platform (02/2015 – 04/2016): Built Weld Annotation Editor and relational model for traceability.

EDUCATION

GRADUATED 2014

Bachelor of Technology (B. Tech), Electrical & Electronics Engineering
Cochin University of Science and Technology (CUSAT)

KEY PROJECTS

RAG Pipeline & LLM Systems, Robogebra – Product Development | April 2025 – Feb 2026

- Built a Retrieval Augmented Generation (RAG) pipeline using LangChain, ChromaDB, and OpenAI embeddings to reduce LLM hallucination and ground responses in proprietary knowledge bases.
- Engineered real time LLM streaming with Server-Sent Events (SSE), delivering token by token responses and enabling live tool execution visibility.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate - AWS, 01/2024

ADDITIONAL HIGHLIGHTS

- Remote delivery experience across the US, Canada, Denmark, and Australia; async collaboration and cross-timezone ownership.
- Independent consultation (Feb 2026 – Present).